

ALIAH UNIVERSITY

Even semester examination, May, 2024

1st year (2nd sem) B .Tech (Mechanical Engineering)

Workshop Practice (MENUGES02)

FULL MARKS: 40

TIME : 1 hr.

1. Answer the all questions

1*10 = 10

A. Angle of a Prick punch..

i) 90° ii) 45° iii) 60° iv) 30°

B. are used on grub screws and socket head cap screws

i) Screw driver ii) Spanner ii) Allen keys iv) Tap wrench

C. are used to make external threads on a rods.

i) Taps ii) Dies iii) Chisels iv) Files

D. is used for permanent marking.

i) Divider ii) Punch iii) Scriber iv) Steel rule.

E. It can be also be used for making perpendicular lines onto a workpiece.

i) Scriber ii) Engineering square iii) Inside caliper iv) Sleeve gauge.

F. It is used to strike the soft materials without damaging the surface.

i) Ball peen hammer ii) Mallet iii) Crosspeen hammer iv) Files

G. Files are made from..

i) Brass ii) Plain Steel iii) High carbon steel iv) Cast iron.

H. is used in finding the center of a cylinder and determining whether a plane's surface is truly horizontal.

i) Angle plate ii) Surface table iii) V- Block iv) Surface gauge

I. Spanners that have a moving jaw to fit different bolt/nut sizes.

i) Combination spanners. ii) Ring spanners. iii) Adjustable spanners
iv) Open-ended spanners.

J are used in metal working to mark lines on work pieces prior to manufacturing.

i) Punches ii) Drills iii) Scribes iv) Letter stamps.

2. Write short note (Any four questions) $2.5 \times 4 = 10$

i) Surface gauge ii) Hacksaw iii) Bench vice iv) Taps & dies v)
Calipers.

3. Answer the following questions. (Any four) $5 \times 4 = 20$

A. Draw a vernier caliper and show a measurement 19.64 mm.

B. Briefly describe constructional parts of outside micrometer.

C. What is positive error? What precaution should be taken while using micrometer?

D. Describe types of files.

E. Draw a schematic view of Vernier Height Gauge with its different parts.

F. Write 10 general safety rules in Mechanical Workshop

Aliah University

End-Semester Examination - 2024

(For B.Tech in Mechanical)-1st Year 2nd Semester

Paper Name: Programming for Problem Solving
Paper Code: CSEUGES01

Full Marks: 80

Time: 3 Hrs

Group – A

(10X1=10)

1. Answer the following (in short any ten(10))

- i. Define local and global variable.
- ii. What is the difference between declaring a variable and defining a variable.
- iii. What is a bug.
- iv. What is the difference between ++i and i++.
- v. What do getchar() and putchar() functions do.
- vi. What is the difference between break and continue statement.
- vii. What is the importance of #include.
- viii. What is dangling pointer.
- ix. Write one difference between character array and string.
- x. What is ASCII.
- xi. What is pseudocode.
- xii. What is the output of the following program.

```
#include<stdio.h>
main()
{
    unsigned int num = 65000;
    while(num++!=0);
    printf("%d",num);
}
```

- xiii. Find the error in the following program.

```
#include<stdio.h>
void main()
{
    int i;
    printf("Enter the value of i");
    scanf("%d",&i);
    If(i<7)
    {
        i++;
        continue;
    }
    printf("Final value of i is: %d",i);
    getch();
}
```

Group – B
(Answer any 6 questions)

(6X5=30)

2. What is a FILE? What are the operations there using a FILE. 1+4
3. What are the differences between a flowchart and an algorithm. 5
4. Write a program to print the first 10 prime numbers using C. Write an algorithm to check a number is prime. 2+3
5. What is array. Write difference between an array and a structure. 1+4
6. What are the symbols we use for making a flowchart. Write a flowchart to find the maximum of three given numbers. 2+3
7. Write the difference between if loop and nested if loop. Write a C program using a switch case to build a system of grading for the students of a school? 2+3
8. What is a pointer? Define dangling pointer. Write an example. 2+2+1
9. Define Dynamic Memory Allocation. What is the meaning of calling malloc(0). 4+1

Group – C
(Answer any 4 questions)

(4X10=40)

10. a) Write the difference between malloc() and calloc(). Write a program to check whether a number is palindrome or not? 4+6
 11. a) What are the storage class specifier for a variable. What are the storage classes, Explain each in brief in the context of place of storage, scope, lifetime and default value? 1+2+7
 12. Define recursion. Give an example of a recursive function. How can recursion be implemented using a program? 2+2+6
 13. Write different types of control statement. Write a C program to demonstrate the use of for() loop? 2+8
 14. What is a pointer. Where is the pointer stored. Define NULL pointer. Implement array of pointer? 1+2+2+5
 15. Write short notes on the following (any two). 5X2 = 10
 - (a) Flowchart
 - (b) Array
 - (c) Function
 - (d) Local and Global variable
-

ALIAH UNIVERSITY
Even Semester Examination (Spring) 2023-'24
PAPER – MATUGBS02
(Engineering Mathematics-II)
Semester – II

Full Marks – 80

Time Allowed– 3 Hours

Unless otherwise stated, symbols have their usual meanings. Use separate answer script for each group.

Group-A (40 Marks)

Answer any FOUR of the following questions:

(4 × 10 = 40)

1. (a) If A is a skew symmetric matrix, prove that $(I - A)(I + A)^{-1}$ is orthogonal. [5]

- (b) If $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, show that $A^2 - 4A - 5I_2 = O$. Hence find A^{-1} . [5]

2. (a) What do you mean by the linear dependence of vectors? Find the values of k , so that the vectors $(1, 2, 0)$, $(k, 0, 1)$, $(1, k, 1)$ are linearly dependent. [1+4]

- (b) Define eigen value and eigen vector of a square matrix. Find the eigen values of the following

matrix $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. [5]

3. (a) Find the Echelon form of the matrix $\begin{bmatrix} 1 & 3 & 2 & 4 & 1 \\ 0 & 0 & 2 & 2 & 0 \\ 2 & 6 & 2 & 6 & 2 \\ 3 & 9 & 1 & 10 & 6 \end{bmatrix}$ using elementary row operations. [5]

- (b) Define rank of a matrix. Find the rank of the matrix $\begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$. [1+4]

4. (a) For what values of k , the following equations $x + y + z = 1$, $2x + y + 4z = k$, $4x + y + 10z = k^2$ have solutions and solve them completely in each case. [5]

- (b) State Cayley-Hamilton theorem. [1]

- (c) (i) Using Cayley-Hamilton theorem, find the inverse of the matrix $\begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. [4]

OR, (ii) If $\hat{A} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, find A^{100} by Cayley-Hamilton theorem. [4]

5. (a) State the conditions for consistency of a non-homogeneous system of equations. Solve the system of equations $x + 5y = 1$, $2y - z = 2$, $x + 3z = -1$. [1+4]

- (b) What do you mean by the linear independence of vectors? Check whether the vectors $(2, 1, 4)$, $(1, -1, 2)$, $(3, 1, -2)$ are linearly independent or not. [1+4]

6. (a) (i) Find the modulus and the principal value of argument of $1 + i \tan \frac{3\pi}{5}$. [6]
 OR, (ii) Show that for any four non-negative real numbers a, b, c and d , the following relation holds $a^5 + b^5 + c^5 + d^5 \geq abcd(a + b + c + d)$. [6]
 (b) Solve the equation $z - 12 = i(9 - 2\bar{z})$, where $z = x + iy$. [4]
7. (a) (i) Find the value of ω^n , where ω is a root of the equation $z^2 + z + 1 = 0$ and $n \in \mathbb{Z}$. [5]
 OR, (ii) Solve $z^3 = 1$. [5]
- (b) (i) Prove that the determinant $\begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$ is a perfect square. [5]
 OR, (ii) If one of the roots of the equation $x^3 - 5x^2 + ax + b = 0$ (where $a, b \in \mathbb{R}$) is $2 + i$, then find other two roots and also find the value of a and b . [5]

GROUP - B (40 marks)

Answer any EIGHT of the following questions:

(8 × 5 = 40)

1. Solve the differential equation $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$. [5]
2. Solve the linear differential equation
 (a) $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{1}{x^2}$, OR, (b) $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$. [5]
3. Solve the Bernoulli's differential equation $\frac{dy}{dx} + \frac{y}{2} = -e^{-x}y^3$. [5]
4. (a) Find an integrating factor (I.F.) of the differential equation $(y \log x + y)dx + dy = 0$ and hence solve the differential equation, [5]
 OR, (b) Find an integrating factor (I.F.) of the differential equation $3ydx - 2xdy + x^2y^{-1}(10ydx - 6xdy) = 0$ and hence solve the differential equation, [5]
5. Solve: $y + px = p^2x^4$, where $p = \frac{dy}{dx}$. (You may use 'solvable for y' method.) [5]
6. Solve: $x^2p^2 + 2pxy + y^2 - 1 = 0$, where $p = \frac{dy}{dx}$. [5]
7. Solve: $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = 3x^2 + 1$. [5]
8. Solve: $\frac{d^2y}{dx^2} - y = xe^x$. [5]
9. Solve: $\frac{d^2y}{dx^2} + a^2y = \cos ax$. [5]
10. (a) Show that the solution of the differential equation (Clairaut's form) $y = px + f(p)$ is $y = cx + f(c)$ where c is an arbitrary constant and hence solve $y = px + p^2$, [5]
 OR, (b) Prove that the necessary and sufficient condition of exactness of the differential equation $Mdx + Ndy = 0$ is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. [5]

— End —

ALIAH UNIVERSITY

Name of the Examination: Even Semester
Subject: Engineering Chemistry
Duration: 3 Hrs

Year: 2024
Course code: CHMUGBS01
Full Marks: 80

Use separate answer scripts for each group

Group A

01. Explain the following terms (answer any FOUR): (a) State function (b) Reversible process (c) Molar heat capacity (d) Entropy (e) Intensive property (f) Joule Thomson phenomenon. 2.5×4
- answer any ALL from the following questions*
02. (a) Deduce the formula for the work done during an isothermal and reversible expansion of an ideal gas. (b) Sketch the pressure-volume (P vs V) graphs for the isothermal expansion of an ideal gas, illustrating both reversible and irreversible processes. 2.5+2.5
03. (a) Derive the relationship $\Delta H = \Delta U + \Delta nRT$ (b) What is the basic principle of Joule-Thomson effect? 2.5+2.5
04. (a) Distinguish between closed and isolated system. Give examples. (b) Differentiate between reversible and irreversible processes. 2.5+2.5
05. (a) What do you understand by C_p and C_v of gases? (b) Why is the value of C_p always greater than that of C_v ? How are they related? 2.5+2.5

Group B (Answer any TEN)

06. Write short notes on:
(i) Teflon (ii) Bakelite
07. How the following polymers prepared? Mention their uses
(i) PVC (ii) LDPE
08. (a) Write informative notes on the mechanism of radical polymerisation
(b) What is meant by green chemistry?
09. (a) What is meant by degree of polymerisation?
(b) Distinguish between addition and condensation polymerisation with examples.
10. (a) How is the Crepe rubber obtained from latex?
(b) Give the structural unit of gutta percha.
11. Explain the following with suitable reactions (i) Polystyrene (ii) Phenol formaldehyde resin
12. Why are silicones called inorganic polymers? Discuss the synthetic procedure of linear chain silicones.
13. What is meant by the vulcanisation of rubber? How vulcanisation takes place? Write its uses?
14. What is Biopolymers? How is it prepared? Give some importance of Biopolymers.
15. (a) How will you prepare Styrene-Butadiene Rubber (SBR). Mention its uses.
(b) What is Ziegler Natta catalyst with some application.
16. Write a note on conducting polymers with reference to oxidative or p-doping with mechanism.
17. Write note on: (a) Compounding of rubber (ii) Conducting polyaniline
18. (a) Explain the importance of green chemistry.
(b) Discuss the use of microwave radiation in green chemistry with an example.
19. (a) Why water is used an alternative solvent in green synthesis.
(b) Prepare aspirin from salicylic acid using green synthesis process.

ENGUGHU01: Communicative English

End Semester Examination 2024

(For the 2nd Semester Students of all the Departments of the Engineering Faculty)

Full Marks: 80

Time: 3 Hours

Section I

1. Answer **ANY TEN** of the following questions:

2x10=20 Marks

1. What is extra-personal communication? ✓
2. What does the word 'Communicare' mean? ✓
3. What is feedback in communication? ✓
4. What is *lingua franca*? ✓
5. What is semantic gap? ✓
6. What is passive listening? ✓
7. What is skimming in reading? ✓
8. What is modal verb? Give an example.
9. Give an example of Grapevine communication. ✓
10. What is active listening? ✓
11. How many consonant sounds are there in English language? ✓
12. What is homophone? Give an example. ✓
13. What is phrase? Give an example. ✓
14. What are the channels of communication?
15. What is monologue?

(Contd.)

Section II

2. Answer ANY FOUR of the following questions: (word limit 200)

5x4=20 Marks

1. Distinguish between formal and informal communication. ✓
2. Define body language. Mention some of the gestures we use to communicate. ✓
3. Write a short dialogue between two friends discussing the price hike.
4. What is the difference between summary and paraphrase? ✓
5. What are the important features of close reading?
6. What are the processes of documenting? ✓
7. What is business communication? What are the methods of business communication? ✓

Section III

3. Answer ANY FOUR of the following questions. (word limit 400)

10X4=40 Marks

- ✓ 1. Identify the notable barriers of an Effective Communication. Discuss the means to overcome those barriers. (6+4)
2. Attaching a Resume, write a job application for the post of a teacher in a college. (10)
3. Define and discuss the differences between Verbal Communication and Non-verbal Communication. (10)
4. Explain the different flows of communication in the business organisations. (10)
5. Define Intra-personal Communication and Inter-personal Communication. Discuss the differences between Intra-personal Communication and Inter-personal Communication. (5+5)
6. Write an email to the Head of your Department to extend the date of the submission of the tutorial project due to your illness. ✓
7. Write a report of the Annual Sports day celebrated in your university.

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AIMM UNIVERSITY
Department of Electrical Engineering, B-Tech
Even (Spring) Semester Examination, 2024 (Reg./Supple)
Subject: Basic Electrical Engineering [EENUGES01]

Full Marks: 80
Year: I

Time: 3 Hours
Semester: II

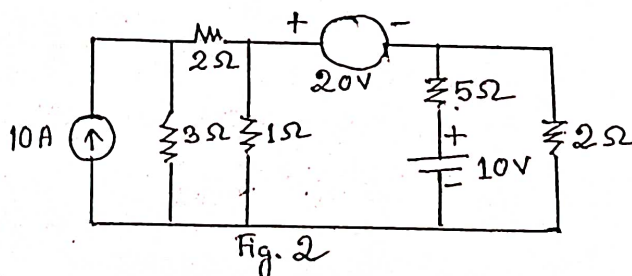
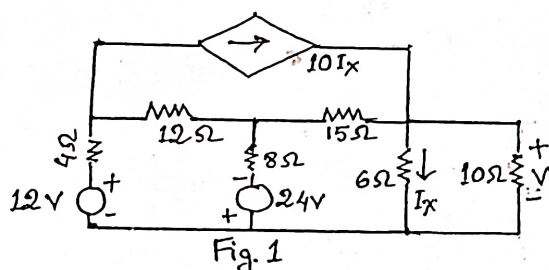
- Instructions:**
1. Mention the question number clearly and write all the parts of a question at one place.
 2. Draw circuit/figure/phasor diagram wherever applicable (including numerical).
 3. Make suitable assumptions wherever necessary, symbols and notations have their usual meanings.

1. i) What is the time constant of series RC circuit with $R=10k\Omega$ and $C=40\mu F$. 2x10= 20 Marks
 ii) Voltage across capacitor and current through an inductor cannot change instantaneously. Justify.
 iii) A 200/50V, 50 Hz single phase transformer is connected to a 200V, 50 Hz supply with secondary winding open. What is the value of maximum flux through the core, if the primary winding has 400 turns?
 iv) Give two basic speed control schemes of DC shunt motor?
 v) When we switch on the lights one after the other in our house, what is the effect on the resistance of the electric circuit of the house? What will happen on the current flowing in the main circuit?
 vi) Is Ohm's law true for all conductors? Name two types of commercially available resistors.
 vii) A resistor of 8Ω and a capacitor of $178\mu F$ are connected in series to a 250V, 50 Hz supply. Net impedance of the circuit is given by 19.6Ω . What is the value of reactive power?
 viii) Can DC be applied to transformers?
 ix) What will happen if DC shunt motor is connected across AC supply?
 x) Give application of all types of DC generators.

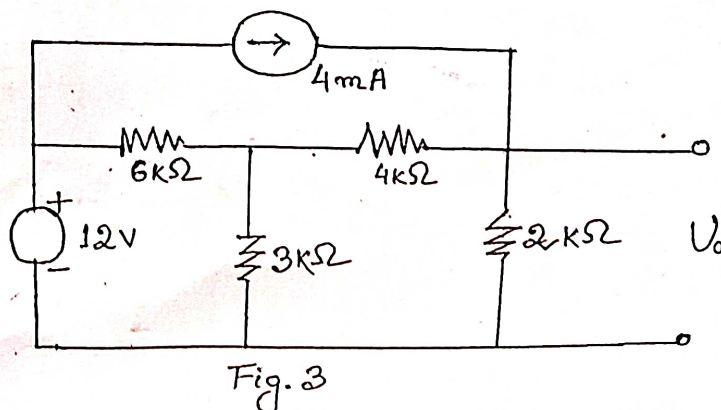
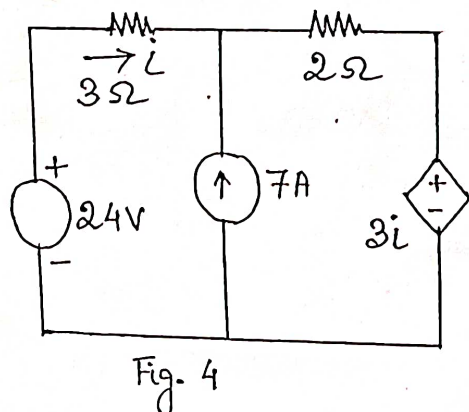
Answer any five questions

5x12=60 Marks

2. a) Use mesh analysis to compute the voltage across 10Ω resistor in Fig. 1.
 b) Use nodal analysis for computing current through the 5Ω resistor in Fig. 2 6+6



3. a) Find V_0 in the circuit of Fig.3 using Thevenin's theorem.
 b) Find the current i using superposition theorem for the circuit shown in Fig.4. 6+6



4. a) In R-C circuit with DC excitation (source free) determine capacitor voltage and capacitor current.
b) In an RC circuit with $R = 25 \Omega$ and $C = 50 \mu F$ is a capacitor being charged with initial zero voltage. What is the time taken for the capacitor voltage to reach 40 % of its steady state value? 6+6
5. a) Write a short note on characteristics of DC series motor. 6+6
b) An 8-pole dc motor has a wave-wound armature with 900 conductors. The useful flux per pole is 25 mWb. Determine the torque exerted when a current of 30A flows in each armature conductor.
6. a) At what condition transformer will give maximum efficiency? Prove for the same. 6+6
b) A 50 kVA single-phase transformer is connected to a 2500V, 50 Hz supply. The primary and secondary turns are 500 and 100 respectively. Determine the following:
i) Open circuit secondary voltage.
ii) Current following through the secondary windings on full-load.
iii) The maximum value of flux.
7. a) A circuit consisting of resistance of 10Ω , an inductance of 13 mH and a capacitance of $150 \mu F$ connected in series. A supply of 100 V at 50 Hz is given to the circuit. Find the impedance, current, power factor and power consumed in the circuit.
b) Write a short note on power triangle. 6+6
8. a) What is an ideal transformer? Draw phasor diagram at different loads for an ideal transformer.
b) Draw R-L-C series circuit with its phasor diagram. Find out the expression of current, phase angle and power. 6+6